Problem-Solution Fit

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Problem-solution fit checks that the solution genuinely addresses the problems found in the earlier phases, for the people who actually have those problems.

1. Fit Table

Problem	Who has it	Solution feature	How it fits
Roll call wastes 5-10 min per lecture	Lecturers, students	Passive presence-based attendance	Attendance accumulates automatically while the lecture runs
Proxy attendance and buddy punching	Institution	Face recognition + presence ratio over the session	A student must be visibly present across the lecture, not just at one moment
No objective attention data	Lecturers	Fused engagement score (gaze, pose, presence, audio)	Gives a live class and per-student index with HIGH / MEDIUM / LOW tiers
Teachers do not know how to react	Lecturers	Rule-based recommendations	Five rules with priorities and concrete action steps
Cloud privacy concerns	Students	100% offline processing; consent-gated enrollment	No video or biometrics leave the laptop; unconsented people stay anonymous
Mistakes in automatic records	Lecturers, students	Manual override with an audit reason	Human can correct the system; reason is stored

2. Customer Segments and Needs

Segment	Core need	What they value in the solution
University lecturers	Save time and see class attention	Automatic attendance, live dashboard, teaching tips
Students	Fair treatment and privacy	Consent, anonymity, smoothing, ability to be corrected
Administrators / departments	Accurate, auditable records	CSV export, session history, override audit trail

3. Existing Alternatives and How This Compares

Alternative	Limitation	This solution
Manual roll call	Slow, interrupts teaching	Runs in the background
Paper sign-in sheet	Easy to fake	Requires visual presence over time
RFID / card tap	Cards can be tapped by friends; needs hardware	No extra hardware beyond a camera
Cloud face-recognition products	Upload video and biometrics; recurring cost	Fully local and free of cloud fees

4. Fit Assessment

The solution addresses every problem statement defined earlier. Its fit is strongest for lecturers who want time savings and engagement insight with strong privacy. The main limit is camera coverage: a single webcam suits a classroom or small hall, while large auditoriums need more cameras (planned future work). Engagement is a heuristic of gaze, pose, and sound, and should be read as a guide, not as proof of understanding.